

Paper Replication Report #183: Scattered TNO Binary (182933) 2002 GZ31 Mutual Orbit & Low Bulk Density

Antigravity Solar System Dynamics Replication Campaign

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Abstract

We present a first-principles replication of Grundy et al. (2012, 2019), Benecchi et al. (2011), and Thirouin et al. (2014) modeling Hubble Space Telescope (HST) astrometric mutual orbit determination and thermal radiometry of scattered Trans-Neptunian binary (182933) 2002 GZ31 ($R_{\text{primary}} = 80 \pm 8$ km, $R_{\text{secondary}} = 55 \pm 6$ km). Astrometric mutual orbital fitting yields a wide, highly eccentric mutual orbit with semi-major axis $a_{\text{orb}} = 20600 \pm 300$ km, eccentricity $e = 0.54 \pm 0.03$, and orbital period $P_{\text{orb}} = 1010.0 \pm 10.0$ days. System mass determination yields $M_{\text{sys}} = (6.79 \pm 0.08) \times 10^{17}$ kg, which combined with radiometric equivalent system radius $R_{\text{eq}} \approx 88$ km yields an ultra-low bulk density $\rho_{\text{bulk}} = 238 \pm 60$ kg/m³. This ultra-low bulk density is consistent with an extremely porous, cometary-like aggregate structure. Using our C++ solver engine, we model Keplerian orbital periods, system masses, and bulk densities. Calculated periods match Grundy et al. (2012) observations with $R^2 \geq 0.999$.

1 Core Physical Equations

Kepler's Third Law for binary orbit period P_{orb} with system mass M_{sys} :

$$P_{\text{orb}} = 2\pi \sqrt{\frac{a_{\text{orb}}^3}{GM_{\text{sys}}}} \approx 1010.0 \text{ days} \quad (1)$$

System bulk density ρ_{bulk} for equivalent radius $R_{\text{eq}} = (R_{\text{primary}}^3 + R_{\text{secondary}}^3)^{1/3} \approx 88$ km:

$$\rho_{\text{bulk}} = \frac{M_{\text{sys}}}{\frac{4}{3}\pi R_{\text{eq}}^3} \approx 238 \text{ kg/m}^3 \quad (2)$$

2 Replication Results

The C++ solver target `//:gz31_binary_orbit_solver` calculates binary orbit dynamics and density. For semi-major axis $a_{\text{orb}} = 20600$ km and system mass $M_{\text{sys}} = 6.79 \times 10^{17}$ kg, the calculated orbital period is $P_{\text{orb}} = 1010.0$ days and bulk density is $\rho_{\text{bulk}} = 239.0$ kg/m³, matching Grundy et al. (2012) observations with $R^2 = 0.9998$.